

hw 0511

14.2

(18) $n=200$ $E = \frac{200}{3} = 66.67$

1) H_0 : 3個入口的 chance 比例相同
 .. 不完全相同

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= \frac{(82-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67}$$

$$= 6.19$$

$df = 3-1=2$

$p\text{-value} = P(\chi^2 > 6.19) = 0.0453 < 0.05$
 $\rightarrow$ reject H_0

(5) 入口 1 比例的 95% 信賴區間

$\hat{p} = \frac{83}{200} = 0.415$

$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.415 \pm 1.96 \sqrt{\frac{0.415(0.585)}{200}}$
 $= (0.3467, 0.4833)$

(22)

H_0 : 每天發生機率相同

H_a : 至少一天不同

$\alpha = 0.05$

$df = 7-1=6$

$E = np = 200 \cdot \frac{1}{7} = 28.57$

$\chi^2 = \sum \frac{(O-E)^2}{E} = 7.63$

$p\text{-value} = 0.7 > 0.05$ non-reject H_0

14.3

(18)

H_0 : 事故致死, 車型大小 ind.

H_a : 有關

$\alpha = 0.05$

$df = (3-1)(2-1) = 2$

	O	E	O-E	$\frac{(O-E)^2}{E}$
Fatal Small	67			0.505
TM	26			0.146
FL	18			0.638
Non-Fatal S	128			0.232
NFM	63			0.088
NFL	46			0.294
				$\sum = 1.885$

Observed Table:

	Small	Medium	Large
Fatal	67	26	18
Non-fatal	128	63	46

Expected:

	Small	Medium	Large
Fatal	61.43	28.04	19.53
Non-fatal	133.57	60.96	42.47

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 1.885$$

$$p\text{-value} = P(\chi^2_{3,0.05} > 0.3895) > 0.05$$

Non-reject H_0

a. H_0 : 薪資, 工作天數獨立
 H_a : 兩者有關

$$\alpha = 0.05$$

$$df = 3 - 2 = 1$$

$$E_{11} = \frac{16 \times 68}{200} = 36.27$$

$$\chi^2 = 5.98$$

$$\chi^2_{0.05, 1} = 12.592$$

Non-reject H_0

$$b. 5.348 < 5.98 < 9.841$$

$$0.5 < p < 0.25$$

$$0.25 < p < 0.5$$

$p > 0.5$, fail to reject H_0